

ORIGINAL 5×3 SLOT MATHEMATICS PROJECT

AETHER LOOM

Portfolio / Demonstration PAR Sheet

Physical reel strips · 243 ways · persistent Markov state · finite bonus reward process

Exact total RTP
96.566915%

Exact feature frequency
1 in 237.74

10M-spin observed RTP
96.638977%

Exact maximum cycle
11,227.60×

Demonstration notice

This document describes a fictional-credit portfolio game. It is not a regulatory submission, certification, gambling product, or claim of production approval. Exact values are computed from model version 1.0.0; simulation values are identified separately.

1. Game overview

Game	Aether Loom
Format	5 reels × 3 rows; 243 ways
Bet	0.20, 0.50, 1, 2, 5, or 10 fictional credits
Win direction	Consecutive reels from reel 1; longest occurrence per symbol
Original mechanic	Paid-spin outcomes move a persistent Tension state. The state selects the next physical reel set and becomes the bonus starting stage.
Bonus	Weave Ascending: 7 initial free spins, maximum 12; two or more Sparks advance the stage and add one spin.
Maximum intended win	11,227.60× bet, exact reachable game-cycle maximum
Jackpot	None

Intended experience. Rest is the stable default. Two Sparks create visible anticipation by moving the loom to Taut, and a second two-Spark result reaches a brief Overdrive spin. The feature inherits earned tension and increases in volatility through richer Wild strips and multipliers of 1×, 1.75×, and 3.5×. The controlling parameters are the exact two-Spark transition probability, state-specific strip composition, and stage multiplier schedule.

2. Symbols and payable

All pays are multiples of total bet. Wild substitutes for all six regular symbols, has no direct pay, and never appears on reel 1. Spark has no cash pay.

Code	Symbol	3 reels	4 reels	5 reels	Role
A	Crown Spool	0.40	1.20	4.00	High
B	Silk Moth	0.30	0.90	2.50	High
C	Copper Shuttle	0.20	0.60	1.50	Medium
D	Dye Vial	0.15	0.40	1.00	Medium
E	Linen Roll	0.10	0.25	0.60	Low
F	Weaver's Knot	0.07	0.15	0.40	Low
W	Aether Wild	–	–	–	Substitute
X	Spark	–	–	–	Scatter/state

For each paying symbol, matching instances on a reel include that symbol and Wild. Ways equal the product of the matching counts across the consecutive reels. Different symbol wins are added. Reel 1 contains no Wild, preventing an all-Wild starting ambiguity.

3. Reel architecture and restrictions

There are six named reel sets. Every reel is a circular ordered strip with 40 stops. The uniformly sampled stop is the top visible symbol and the following two positions are the middle and bottom. Wild and Spark occurrences are separated by at least three circular positions, so a three-row window contains at most one of each special per reel.

Set	Use	Stops/reel	Restriction
BASE_REST	Paid state 0	40	1 Spark/reel; 0 Wild on R1
BASE_TAUT	Paid state 1	40	1–2 Sparks/reel; 0 Wild on R1
BASE_OVERDRIVE	Paid state 2	40	2 Sparks/reel; 4 Wilds on R2–R5
BONUS_REST	Feature stage 0	40	2 Sparks/reel; 4 Wilds on R2–R5
BONUS_TAUT	Feature stage 1	40	2 Sparks/reel; 6 Wilds on R2–R5
BONUS_OVERDRIVE	Feature stage 2	40	2 Sparks/reel; 8 Wilds on R2–R5

4. Persistent Tension mathematics

After a paid spin: 3+ Sparks trigger the feature and reset to Rest afterward; an Overdrive paid spin vents to Rest; exactly 2 Sparks move Rest→Taut or Taut→Overdrive; 0 Sparks lower tension one step; 1 Spark holds.

Current	Next Rest	Next Taut	Next Overdrive
Rest	95.5481%	4.4519%	0.0000%
Taut	58.0910%	34.0913%	7.8177%
Overdrive	100.0000%	0.0000%	0.0000%
Stationary mix	93.2116%	6.2962%	0.4922%

The transition matrix comes from exact scatter-count polynomials for each active strip set. The stationary vector weights base EV and state-conditioned feature entry in the long-run model.

5. Weave Ascending feature mathematics

The trigger is 3+ Sparks on a paid spin. The feature inherits the triggering paid-spin state, begins with seven spins, and ends after the queue is empty or 12 spins have played. On each feature spin, 2+ Sparks add one spin and advance one stage, capped at Overdrive.

Starting stage	Multiplier	Exact expected award	Exact expected spins
Rest	1.00×	32.431583×	8.370532
Taut	1.75×	61.467090×	8.370532
Overdrive	3.50×	96.017658×	8.370532
Trigger-weighted	—	38.360288×	8.370532

The feature is a finite Markov reward process indexed by current stage, spins remaining, and spins played. Exact stage EV is added to the probability-weighted next-state values. This recursion terminates at the 12-spin bound.

6. RTP allocation

Component	Exact RTP	Method
Base 243-ways wins	80.431642%	exact window-count enumeration
Weave Ascending	16.135273%	exact trigger weighting + reward recursion
Scatter cash awards	0.000000%	rule: no direct pay
Jackpots	0.000000%	not used
Total	96.566915%	component sum

7. Simulation methodology and statistics

The independent Monte Carlo executable ran 10,000,000 paid spins with PCG32 seed 20260908. Each paid cycle includes any triggered feature. Welford's online recurrence accumulates payout mean and variance without storing outcomes. The RTP interval uses sample variance: $\bar{x} \pm 1.96s/\sqrt{N}$.

Total wager	10,000,000	Total payout	9,663,897.70
Observed RTP	96.638977%	95% CI	96.398778–96.879176%
Overall hit frequency	65.109340%	Base hit frequency	64.838740%
Feature frequency	0.420600%	Feature triggers	42,060
Free spins / paid spin	3.520610%	Mean spins / feature	8.370447
Average non-zero award	1.484257×	Free spins played	352,061
Variance	15.018659 bet ²	Standard deviation	3.875392× bet
Maximum observed	731.79×	Exact maximum	11,227.60×

8. Exact versus simulation

Quantity	Exact	Simulation	Error (pp)
Base ways RTP	80.431642%	80.499687%	+0.068046
Feature RTP	16.135273%	16.139290%	+0.004016
Total RTP	96.566915%	96.638977%	+0.072062
Base hit probability	64.820986%	64.838740%	+0.017754
Feature-entry probability	0.420624%	0.420600%	-0.000024

The exact total lies inside the observed 95% interval. Conditional feature award is 38.360288× exact and 38.372063× simulated. Mean feature duration is 8.370532 exact and 8.370447 simulated. No discrepancy was hidden or corrected after simulation.

9. Sensitivity analysis

Each row changes one parameter family; RTP, feature rate, and maximum are recalculated exactly. Hit rate and σ use 500,000 spins per row.

Parameter	Level	Exact RTP	Feature	Sim hit	Sim σ	Exact max
Baseline	configured	96.5669%	0.4206%	65.0204%	3.8526	11,227.60×
Bonus Wilds/reel	-1	93.2367%	0.4206%	65.1110%	3.1828	11,227.60×
Bonus Wilds/reel	+1	100.7503%	0.4206%	65.1908%	4.6621	11,563.60×
Base Spark density	lower	95.3858%	0.0000%	70.0614%	1.5150	11,227.15×
Base Spark density	higher	212.2207%	3.1288%	61.7292%	11.0465	11,247.30×
Stage multipliers	1/1.5/3	94.5155%	0.4206%	65.0882%	3.3200	9,626.60×
Stage multipliers	1/2/4	98.6183%	0.4206%	65.1176%	4.3843	12,828.60×
Initial free spins	6	93.3031%	0.4206%	65.0942%	3.3410	10,058.60×
Initial free spins	8	99.9501%	0.4206%	65.1670%	4.5126	12,396.60×

The high Spark case is a boundary experiment that demonstrates why feature frequency and stationary state occupancy must be designed together; it is not a candidate configuration.

10. Ordered reel strips

Code map: A Crown, B Moth, C Shuttle, D Dye, E Linen, F Knot, W Wild, X Spark. These strings are canonical and circular.

BASE_REST

R1 FEFBCDEBECECAFFBFEECFBDFDFDAXFFDFDFFAE

R2 EEXDFCFABEDEFDWBEBFAFEFADWEEFCDFBFCCFCF

R3 DEBDFCFFAAFFCEBCWEFFEDDBECFADFEXCEDWAFBF

R4 DECBAFAFFCEEFFBEECBDFEDFWBDFCEXBACFDWDF

R5 EDFCFEECFBDFDCEXWECAFCFFAFCDDEEDWDFBFADBF

BASE_TAUT

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R1 DDCDFDFBFCABFEAEFFECFBFFEBDEDEACFFDEFEFX
R2 EDEWDFEFCEECABDEFFBDFDWBFWFFACFDCAEXCBXF
R3 DFACFEFWEDECCDCEBAECBEBFFBFDWFAFDXDAEF
R4 WCFBDFEBWEFBCDAAECXFCFCBFEDXFFEADFFWEDDB
R5 FBFFEFDCDCFBDAEBCACFFDWCFCDEEFWEWFEFEB

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BASE_OVERDRIVE / BONUS_REST

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R1 BDBFFAFXDEADCEDFCFCEFFBEEDXFDBFFEECEFECECF
R2 WCEBXDWBFEFEACFAFABDCDEFEFCBWEFDCWDFDXF
R3 WCBFFCCXDDFADEWXBDFFEDEFAAEBDWFEFCWBCA
R4 CFXBCFDECFCWCADEEFWBDFAEDFBXCFWFEWEDF
R5 WEDWFCBWFECFCBXCFCDCFCFDFAEXFAWEBDADFBE

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BONUS_TAUT

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R1 FXFCFFBBEDAFAEDFDEEDCEFFDCEDFXFFEBBCBAF
R2 FFWXDAAWBEFWFCADBCCEXWEDEECCFWFCFEFDDWBB
R3 EFBWCDBCFEBXAWFEAWCEBWFFAAWDDXFCDFDEFCD
R4 WDFDBFBCBEBFFDWBDAWFAWCEWFCBWEFCFWDXAE
R5 WDCEBBAAWCDBWCFWFDWAXECCDFFEWXXFFBEED

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BONUS_OVERDRIVE

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R1 DFDDAFXCCXFEFCFBCDFFBAEFEEFFCDDEEAFFEBB
R2 BADWDDWDAFDWBFCEWBBFCWEWCFCFEXECCWAFDWF
R3 EXFCAFAFWBBWDFDCAWCEWAEWDBCFWXEDDWFDCW
R4 WFEWCEWBCXEDWEDDACAFAWXCWCDWFBWBFFAFDDBB
R5 EWFXXFBCECDEWDCWECAAFDXDWFBBFWDACWCFBCD

```

Full per-reel count tables are included in [docs/reel-strips.md](#).

11. Version and reproducibility

Game/model version	1.0.0
Configuration hash	SHA-256 prefix 6045448975a0
Canonical config	config/game-config.json
Canonical engine	SlotGame.Core (C# / .NET 8)
Exact analyzer	SlotGame.Core.ExactAnalyzer + ExtremeAnalyzer
Monte Carlo	SlotGame.Simulation; seed 20260908
Report classification	Portfolio / demonstration PAR sheet